CLOUD COMPUTING PRACTICAL MANUAL

Windows-Based Practical: Using PowerShell to Manage Files, Processes and Services

Level: BCA
Platform: Windows OS
Requirement: Windows PowerShell

Purpose of the Practical

This practical introduces students to Windows system administration through PowerShell. It covers file management, process management, Windows services and basic automation. These skills are useful in cloud computing because cloud administrators frequently manage servers and virtual machines through command-line tools and scripts.

Learning Objectives

• Navigate and inspect the Windows file system using PowerShell.

• Create, read, copy, move, rename and delete files and folders.

• View, start and stop Windows processes.

• View, start and stop Windows services.

• Create a simple PowerShell script for backup automation.

• Understand how command-line administration relates to cloud/server administration.

Prerequisites

• Windows 10 or Windows 11.

• Windows PowerShell.

• Basic familiarity with files, folders and Windows.

• Administrator privileges may be required for some service operations.

How to Open PowerShell

1. Click Start.
2. Type **PowerShell**.
3. Open **Windows PowerShell**.
4. For service-management tasks, use **Run as administrator** if required.

Safety Note

> Use a lab/test machine. Do not stop or disable unfamiliar Windows processes or services. The Print Spooler is used as the controlled service example in this manual.

Part A — File Management

1. Check the Current Directory

```
Get-Location
```

Displays the directory in which PowerShell is currently operating.

> Example result: C:\Users\Student.

2. List Files and Folders

```
Get-ChildItem
```

Lists files and subfolders in the current directory.

> Aliases: dir and ls.

3. Create a Folder

```
New-Item -Path "C:\CloudLab" -ItemType Directory
```

Creates a folder named CloudLab.

> Verify using: Get-ChildItem C:\

4. Create a Text File

```
New-Item -Path "C:\CloudLab\student.txt" -ItemType File
```

Creates an empty text file.

> Add text using: "Cloud Computing Practical" | Out-File "C:\CloudLab\student.txt"

5. Read the File

```
Get-Content "C:\CloudLab\student.txt"
```

Displays the file contents.

6. Create a Backup Folder

```
New-Item -Path "C:\CloudLab\Backup" -ItemType Directory
```

Creates the Backup subfolder.

7. Copy a File

```
Copy-Item "C:\CloudLab\student.txt" "C:\CloudLab\Backup\student.txt"
```

Copies the file into Backup.

8. Rename a File

```
Rename-Item "C:\CloudLab\student.txt" "cloud.txt"
```

Changes the file name to cloud.txt.

9. Move a File

```
Move-Item "C:\CloudLab\cloud.txt" "C:\CloudLab\Backup\"
```

Moves cloud.txt into Backup.

10. Delete a File

```
Remove-Item "C:\CloudLab\Backup\cloud.txt"
```

Deletes the selected file.

> Use carefully; it does not move the file to the Recycle Bin.

Quick Reference — File Commands

Command	Purpose
Get-Location	Show current directory
Get-ChildItem	List files/folders
New-Item	Create file/folder
Copy-Item	Copy files/folders
Move-Item	Move files/folders
Rename-Item	Rename
Remove-Item	Delete
Get-Content	Read file contents

Part B — Process Management

11. View Running Processes

```
Get-Process
```

Displays processes currently running on Windows.

12. Find a Particular Process

```
Get-Process chrome
```

Displays Chrome if it is running. Otherwise choose a process shown by Get-Process.

13. Filter Processes by CPU Time

```
Get-Process | Where-Object {$_.CPU -gt 100}
```

Shows processes whose reported CPU time is greater than 100 seconds; results vary by system.

14. Start a Process

```
Start-Process notepad
```

Starts Microsoft Notepad.

15. Stop a Process

```
Get-Process notepad
Stop-Process -Name notepad
```

Identifies and then stops Notepad. Use this only with a process intentionally started for the exercise.

Suggested Activity

• Open Notepad using Start-Process.

• Verify it with Get-Process.

• Inspect it using Get-Process notepad.

• Stop it using Stop-Process -Name notepad.

• Verify it is no longer running.

Part C — Windows Service Management

Services are background programs that provide functions to Windows. Managing services introduces students to server administration.

16. View Windows Services

```
Get-Service
```

Lists services and their current status.

17. Find the Print Spooler

```
Get-Service Spooler
```

Displays the status of the Print Spooler service.

18. Start the Service

```
Start-Service Spooler
```

Starts Print Spooler. Administrator permission may be required.

19. Verify Status

```
Get-Service Spooler
```

Confirms the current service status.

20. Stop the Service

```
Stop-Service Spooler
```

Stops Print Spooler for the controlled lab exercise.

21. Start It Again

```
Start-Service Spooler
```

Restores the service after the exercise.

> Important: Service names vary by Windows edition and installed software. If Spooler is unavailable, use Get-Service to identify an appropriate non-critical service. Do not experiment with critical system services.

Part D — Basic PowerShell Automation

Automation is an important connection between this practical and cloud computing. A script can perform repetitive administrative work consistently with less manual effort.

22. Create a Backup Folder

```
New-Item "C:\CloudLab\Backup" -ItemType Directory -Force
```

The -Force option allows the command to continue if the folder already exists.

23. Create Sample Data

```
"Important student data" | Out-File "C:\CloudLab\data.txt"
```

Creates data.txt and writes sample text into it.

24. Copy Data into Backup

```
Copy-Item "C:\CloudLab\data.txt" "C:\CloudLab\Backup\" -Force
```

Copies the file into Backup; -Force permits overwriting an existing copy.

25. Create the Backup Script

```
New-Item "C:\CloudLab\Backup" -ItemType Directory -Force

Copy-Item "C:\CloudLab\*.txt" "C:\CloudLab\Backup\" -Force

Write-Host "Backup completed successfully."
```

Save these commands as **backup.ps1**. The script creates the Backup folder, copies all .txt files from CloudLab, and displays a completion message.

26. Execute the Script

```
cd "C:\CloudLab"
.\backup.ps1
```

If Windows blocks script execution because of execution policy, the instructor should configure an appropriate policy in the controlled lab environment rather than students changing security settings on their own.

How This Relates to Cloud Computing

Cloud platforms provide virtual machines, storage, networking and server services. Administrators need to inspect resources, manage processes and services, move data and automate repetitive tasks. PowerShell provides these capabilities for Windows environments.

This practical	Cloud/server concept
Manage files	Storage administration
Manage processes	Compute/resource management
Manage services	Server/application administration
PowerShell scripts	Automation
Backup script	Data protection and scheduled operations

Complete Command Reference

```
Get-Location
Get-ChildItem
New-Item
Copy-Item
Move-Item
Rename-Item
Remove-Item
Get-Content
Get-Process
Start-Process
Stop-Process
Get-Service
Start-Service
Stop-Service
```

Suggested Practical Record Format

Aim: To use Windows PowerShell to manage files, processes, services and perform basic automation.

Requirements: Windows 10/11 and Windows PowerShell.

Procedure: Perform Parts A–D and record commands and observed outputs.

Result: Students successfully managed Windows files, processes and services using PowerShell and created a basic automated backup script.

Viva Questions

1. What is PowerShell?

A command-line shell and scripting environment used to manage and automate Windows and other systems.

2. What is the difference between Get-ChildItem and Get-Content?

Get-ChildItem lists files/folders; Get-Content reads a file's contents.

3. What does Get-Process do?

It displays running processes.

4. What is a Windows service?

A background program that provides a system or application function.

5. Why is PowerShell useful in cloud computing?

It allows administrators to manage resources and automate repetitive tasks through commands and scripts.

6. What is automation?

Using scripts or tools to perform tasks automatically with minimal manual intervention.

7. Why might administrator privileges be required?

Some system-level processes and services are protected and can only be managed by authorized users.